Solution Overview

NetApp Solutions

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Solution Overview

NetApp ONTAP AI and Cloud Sync

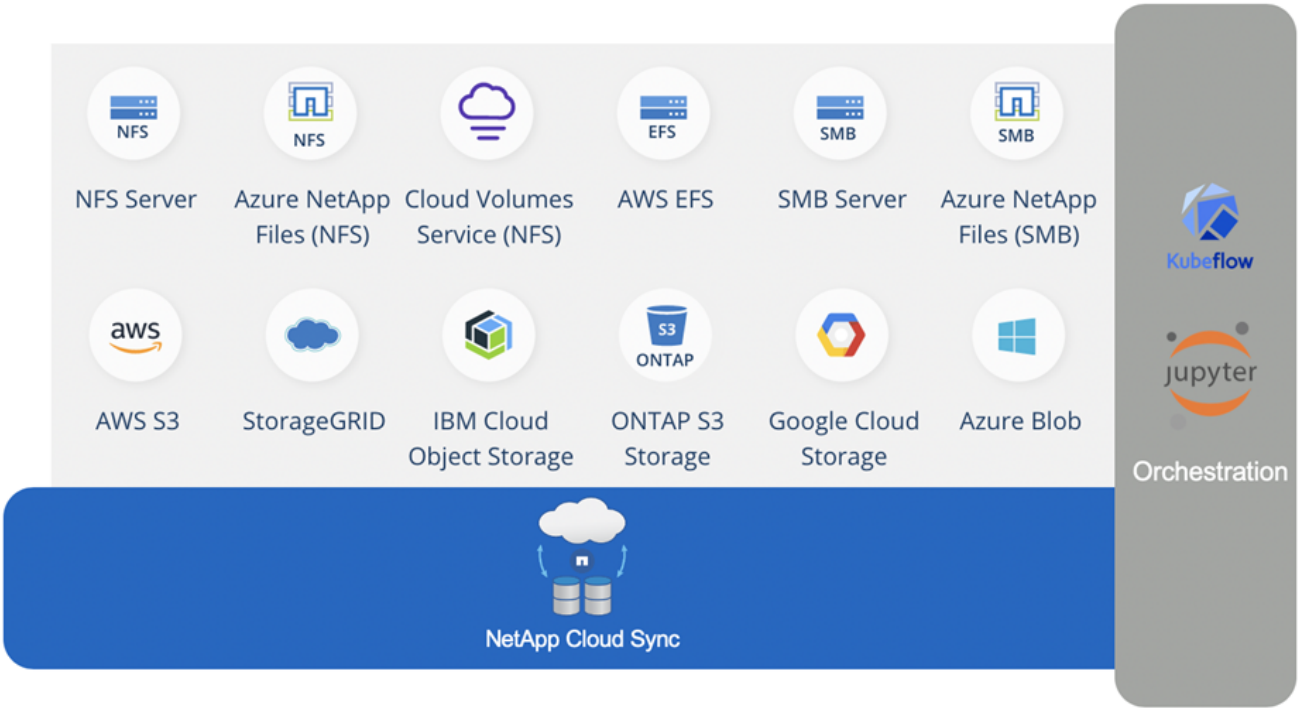
The NetApp ONTAP AI architecture, powered by NVIDIA DGX systems and NetApp cloud-connected storage systems, was developed and verified by NetApp and NVIDIA. This reference architecture gives IT organizations the following advantages:

- Eliminates design complexities
- Enables independent scaling of compute and storage
- Enables customers to start small and scale seamlessly
- Offers a range of storage options for various performance and cost pointsNetApp ONTAP AI tightly integrates DGX systems and NetApp AFF A220 storage systems with state-of-the-art networking. NetApp ONTAP AI and DGX systems simplify AI deployments by eliminating design complexity and guesswork. Customers can start small and grow their systems in an uninterrupted manner while intelligently managing data from the edge to the core to the cloud and back.

NetApp Cloud Sync enables you to move data easily over various protocols, whether it’s between two NFS shares, two CIFS shares, or one file share and Amazon S3, Amazon Elastic File System (EFS), or Azure Blob storage. Active-active operation means that you can continue to work with both source and target at the same time, incrementally synchronizing data changes when required. By enabling you to move and incrementally synchronize data between any source and destination system, whether on-premises or cloud-based, Cloud Sync opens up a wide variety of new ways in which you can use data. Migrating data between on-premises systems, cloud on-boarding and cloud migration, or collaboration and data analytics all become easily achievable. The figure below shows available sources and destinations.

In conversational AI systems, developers can leverage Cloud Sync to archive conversation history from the cloud to data centers to enable offline training of natural language processing (NLP) models. By training models to recognize more intents, the conversational AI system will be better equipped to manage more complex questions from end-users.

NVIDIA Jarvis Multimodal Framework



NVIDIA Jarvis is an end-to-end framework for building conversational AI services. It includes the following GPU-optimized services:

- Automatic speech recognition (ASR)
- Natural language understanding (NLU)
- Integration with domain-specific fulfillment services
- Text-to-speech (TTS)
- Computer vision (CV)Jarvis-based services use state-of-the-art deep learning models to address the complex and challenging task of real-time conversational AI. To enable real-time, natural interaction with an end user, the models need to complete computation in under 300 milliseconds. Natural interactions are challenging, requiring multimodal sensory integration. Model pipelines are also complex and require coordination across the above services.

Jarvis is a fully accelerated, application framework for building multimodal conversational AI services that use an end-to-end deep learning pipeline. The Jarvis framework includes pretrained conversational AI models, tools, and optimized end-to-end services for speech, vision, and NLU tasks. In addition to AI services, Jarvis enables you to fuse vision, audio, and other sensor inputs simultaneously to deliver capabilities such as multi-user, multi-context conversations in applications such as virtual assistants, multi-user diarization, and call center assistants.

NVIDIA NeMo

NVIDIA NeMo is an open-source Python toolkit for building, training, and fine-tuning GPU-accelerated state-of-the-art conversational AI models using easy-to-use application programming interfaces (APIs). NeMo runs mixed precision compute using Tensor Cores in NVIDIA GPUs and can scale up to multiple GPUs easily to deliver the highest training performance possible. NeMo is used to build models for real-time ASR, NLP, and TTS applications such as video call transcriptions, intelligent video assistants, and automated call center support across different industry verticals, including healthcare, finance, retail, and telecommunications.

We used NeMo to train models that recognize complex intents from user questions in archived conversation

history. This training extends the capabilities of the retail virtual assistant beyond what Jarvis supports as delivered.

Retail Use Case Summary

Using NVIDIA Jarvis, we built a virtual retail assistant that accepts speech or text input and answers questions regarding weather, points-of-interest, and inventory pricing. The conversational AI system is able to remember conversation flow, for example, ask a follow-up question if the user does not specify location for weather or points-of-interest. The system also recognizes complex entities such as “Thai food” or “laptop memory.” It understands natural language questions like “will it rain next week in Los Angeles?” A demonstration of the retail virtual assistant can be found in Customize States and Flows for Retail Use Case.

Next: Solution Technology

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